FOODHUB ORDER

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Contents..

1) How many rows and columns are present in the data?
2) What are the datatypes of the different columns in the dataset?
3) Are there any missing values in the data? If yes, treat them using an appropriate method?
4) Check the statistical summary of the data. What is the minimum, average, and maximum time it takes for food to be prepared once an order is placed?
5) How many orders are not rated?
6) Explore all the variables and provide observations on their distributions. (Generally, histograms, boxplots, count plots, etc. are used for univariate exploration.)

   a) Cuisine type distribution

   b) Cost of the order distribution

   c) Day of the week distribution

   d) Rating distribution

   e) Food Preparation time distribution

   f) Delivery time distribution

7) Which are the top 5 restaurants in terms of the number of orders received?

8) Which is the most popular cuisine on weekends?

9) What percentage of the orders cost more than 20 dollars?

10) What is the mean order delivery time?

11) The company has decided to give 20% discount vouchers to the top 3 most frequent customers. Find the IDs of these customers and the number of orders they placed.

12) Perform a multivariate analysis to explore relationships between the important variables in the dataset. (It is a good idea to explore relations between numerical variables as well as relations between numerical and categorical variables)

   a) Cuisine vs Cost of the order relationship

   b) Cuisine vs food preparation time relationship

   c) Day of the week and delivery time relationship

d) Run the below code and write your observations on the revenue generated by the restaurants.

e) Rating vs Delivery time relationship

f) Rating vs food preparation time relationship

g) Rating vs cost of the order relationship

h) Correlation among variables

13) The company wants to provide a promotional offer in the advertisement of the restaurants. The condition to get the offer is that the restaurants must have a rating count of more than 50 and the average rating should be greater than 4. Find the restaurants fulfilling the criteria to get the promotional offer.
14) The company wants to provide a promotional offer in the advertisement of the restaurants. The condition to get the offer is that the restaurants must have a rating count of more than 50 and the average rating should be greater than 4. Find the restaurants fulfilling the criteria to get the promotional offer.
15) The company charges the restaurant 25% on the orders having cost greater than 20 dollars and 15% on the orders having cost greater than 5 dollars. Find the net revenue generated by the company across all orders.
16) The company wants to analyse the total time required to deliver the food. What percentage of orders take more than 60 minutes to get delivered from the time the order is placed? (The food has to be prepared and then delivered.)
17) The company wants to analyse the delivery time of the orders on weekdays and weekends. How does the mean delivery time vary during weekdays and weekends?

1) The dataset has 1898 rows & 9 columns

```
# Check the shape of the dataset
df.shape ## Fill in the blank
```

```
(1898, 9)
```

2) There are 4 numerical ,4 object & 1 float column in the dataset. Rest all the columns have 1898 entries.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1898 entries, 0 to 1897
Data columns (total 9 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   order_id               1898 non-null   int64  
 1   customer_id            1898 non-null   int64  
 2   restaurant_name        1898 non-null   object 
 3   cuisine_type           1898 non-null   object 
 4   cost_of_the_order      1898 non-null   float64
 5   day_of_the_week        1898 non-null   object 
 6   rating                 1898 non-null   object 
 7   food_preparation_time  1898 non-null   int64  
 8   delivery_time          1898 non-null   int64  
dtypes: float64(1), int64(4), object(4)
memory usage: 133.6+ KB
```

3) In below, there are no missing values and all the columns have 1898 non-null observations.

```
# Checking for missing values in the data
df.isnull().value_counts()  #Write the appropriate function to print the sum of null values for each column
```

```
order_id  customer_id  restaurant_name  cuisine_type  cost_of_the_order  day_of_the_week  rating  food_preparation_time  delivery_time
False     False        False            False         False              False            False   False                  False            1898
Name: count, dtype: int64
```

4) Minimum time - 20.0
Average time - 27.37
Maximum time - 35.00

```
# Get the summary statistics of the numerical
df['food_preparation_time'].describe() ## Wri
```

```
count    1898.000000
mean       27.371970
std         4.632481
min        20.000000
25%        23.000000
50%        27.000000
75%        31.000000
max        35.000000
Name: food_preparation_time, dtype: float64
```

5) 736 orders are not rated by the customer.

```python
df['rating'].value_counts() ## Complete the code
```

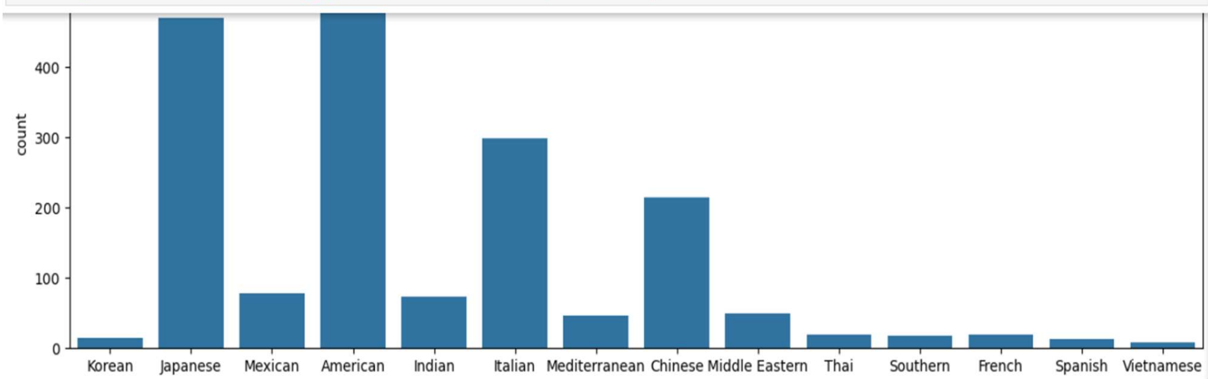
```
rating
Not given    736
5            588
4            386
3            188
Name: count, dtype: int64
```

6)

- Each order placed on the app Order ID and hence 1898 unique order ID’s
- Restaurants registered with the app as service provider is 178

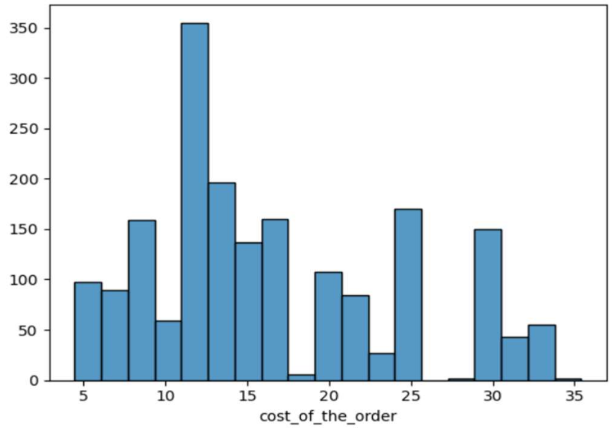
6.1) Cuisine type distribution

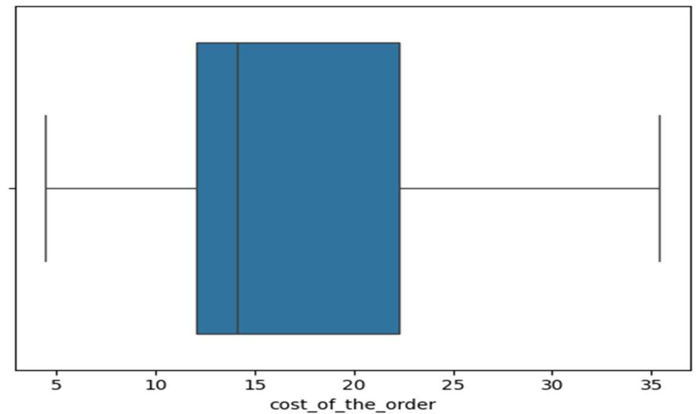
```python
plt.figure(figsize = (15,5))
sns.countplot(data = df, x = 'cuisine_type') ## Create a countplot for cuisine type.
```



- American cuisine has the highest orders followed by Japanese & Italian
- Both Italian and Chinese have a significant number of orders, though far less than American and Japanese.
- Mexican, Indian, Mediterranean and Middle Eastern have moderate counts indicating so its most popular

6.2) Cost of the order distribution



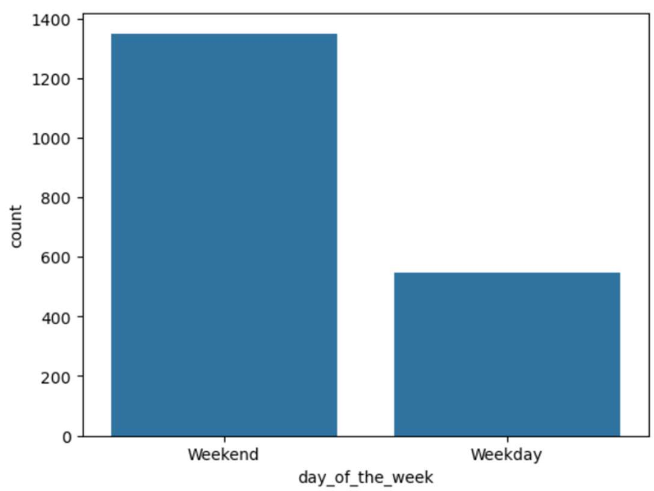


- A large number of orders placed cost between 11 - 17.
- The average cost of the order is also close to 17 as the median is close to 14.

6.3) Day of the week distribution

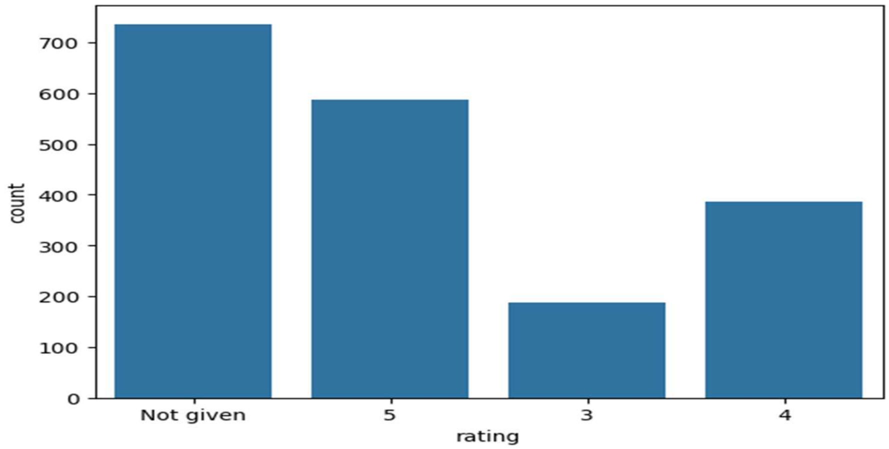
```
]: sns.countplot(data = df, x = 'day_of_the_week') ## Complete the code to plot a bar graph
```

```
]: <Axes: xlabel='day_of_the_week', ylabel='count'>
```



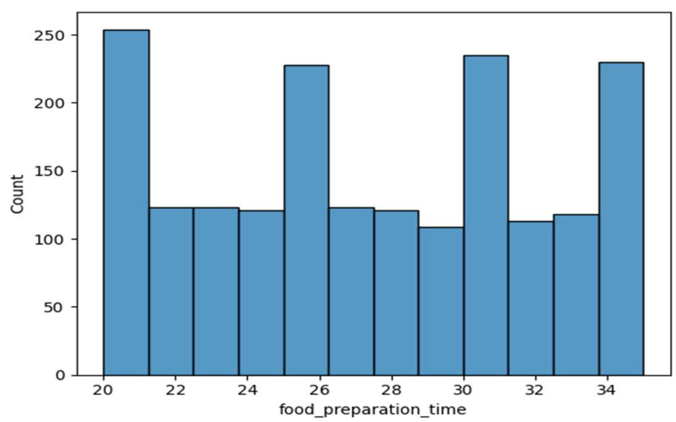
- A large number of orders are placed on weekends
- Order placed on weekday is less than to the orders placed on weekends.

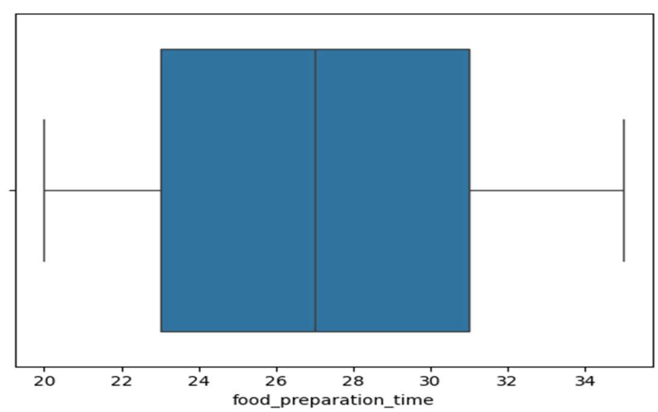
6.4) Rating distribution



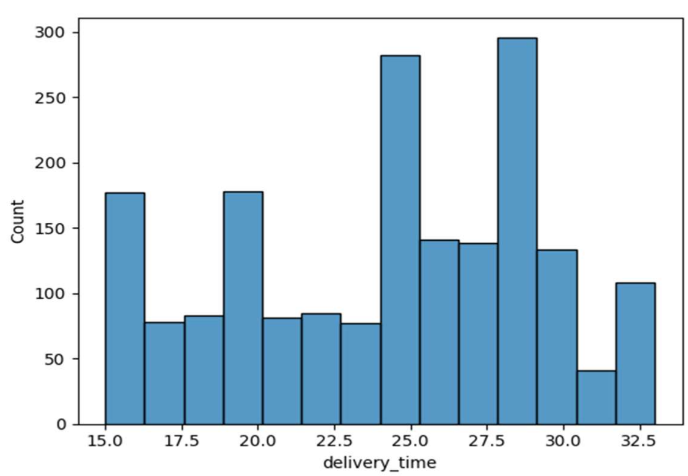
- Approximately 40% customers did not share a rating.

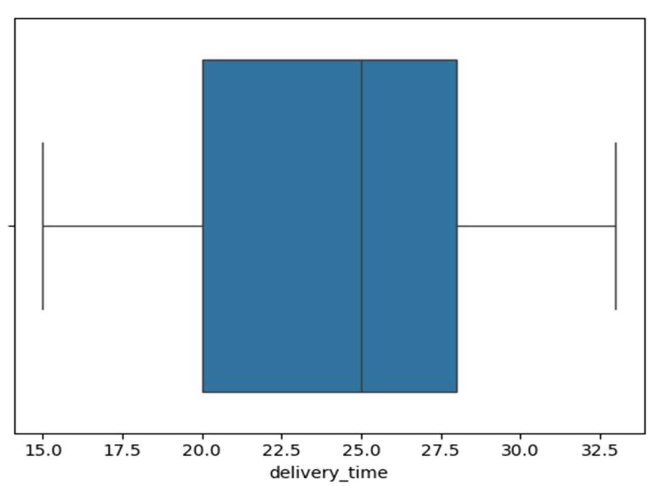
6.5) Food Preparation time distribution





6.6) Delivery time distribution





5) The top 5 restaurants are 'Shake Shack', 'The Meatball shop', 'Blue Ribbon Sushi', 'Blue Ribbon Fried Chicken' and 'Parm'.

```
# Get top 5 restaurants with highest number of orders
df.groupby('restaurant_name')['order_id'].count().sort_values(ascending=False).head(5)
```

```
restaurant_name
Shake Shack                  219
The Meatball Shop            132
Blue Ribbon Sushi            119
Blue Ribbon Fried Chicken     96
Parm                          68
Name: order_id, dtype: int64
```

6) American followed by Japanese and then Italian.

```
<bound method Series.idxmax of cuisine_type
American          415
Japanese          335
Italian           207
Chinese           163
Mexican            53
Indian             49
Mediterranean      32
Middle Eastern     32
Thai               15
French             13
Korean             11
Southern           11
Spanish            11
Vietnamese          4
Name: count, dtype: int64>
```

7) The percentage of orders is above then 20 dollars-29.24%

```
# Get orders that cost above 20 dollars
df_greater_than_20 = df[df['cost_of_the_order']>20] ## Write the appropriate column name to get

# Calculate the number of total orders where the cost is above 20 dollars
print('The number of total orders that cost above 20 dollars is:', df_greater_than_20.shape[0])

# Calculate percentage of such orders in the dataset
percentage = (df_greater_than_20.shape[0] / df.shape[0]) * 100

print("Percentage of orders above 20 dollars:", round(percentage, 2), '%')
```

```
The number of total orders that cost above 20 dollars is: 555
Percentage of orders above 20 dollars: 29.24 %
```

8) The mean delivery time of this dataset is 24.16 min

```python
# Get the mean delivery time
mean_del_time = df['delivery_time'].mean()  ## Write the appropriate function to obtain the mean delivery time

print('The mean delivery time for this dataset is', round(mean_del_time, 2), 'minutes')
```

```
The mean delivery time for this dataset is 24.16 minutes
```

9) There are the top 3 customer id with their order count

```python
# Get the counts of  each customer_id
df['customer_id'].value_counts().head(3)
```

```
customer_id
52832    13
47440    10
83287     9
Name: count, dtype: int64
```

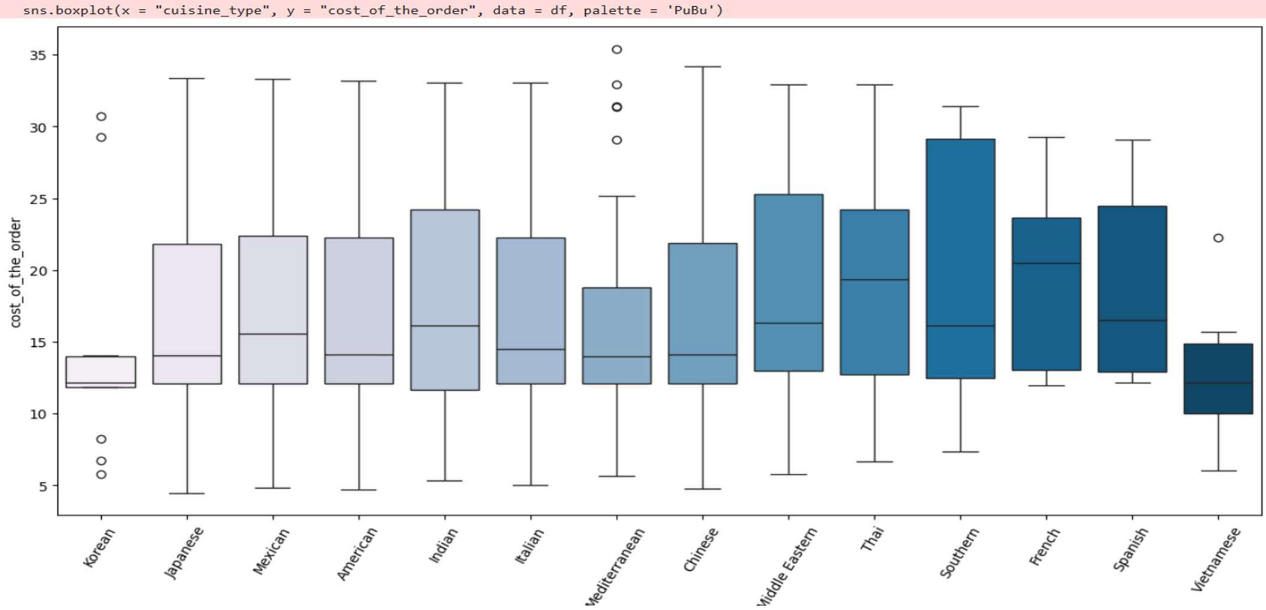
10)

12.1) Cuisine vs Cost of the order relationship

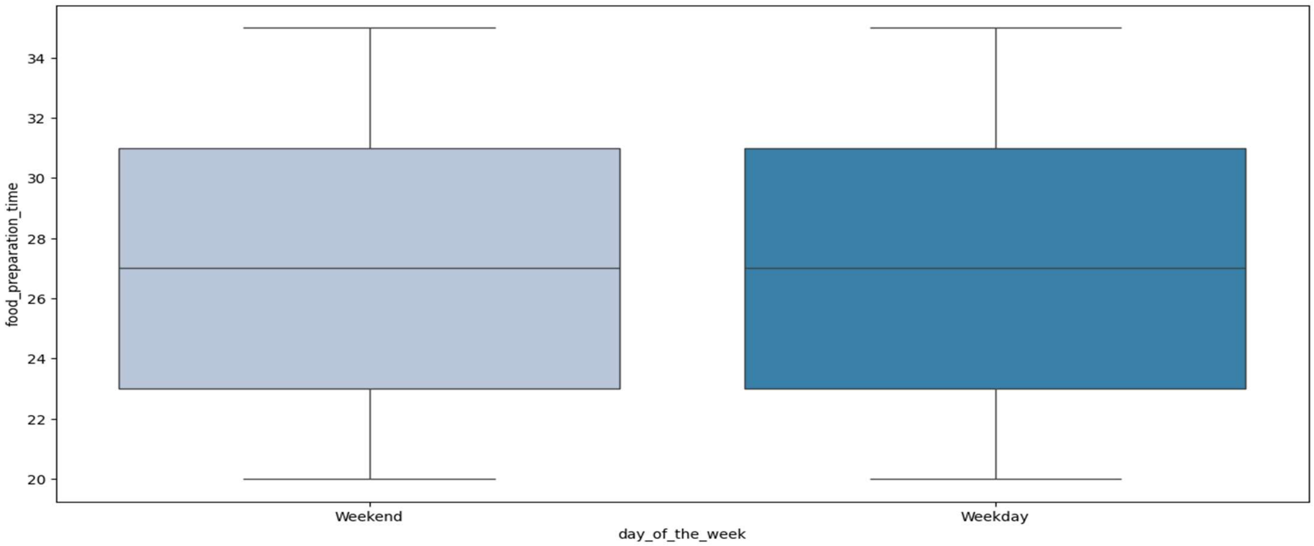
- The max cost of the order for various cuisine types, ranges between 25-35.

12.2) Cuisine vs food preparation time relationship

```
ame effect.
  sns.boxplot(x = "cuisine_type", y = "cost_of_the_order", data = df, palette = 'PuBu')
```



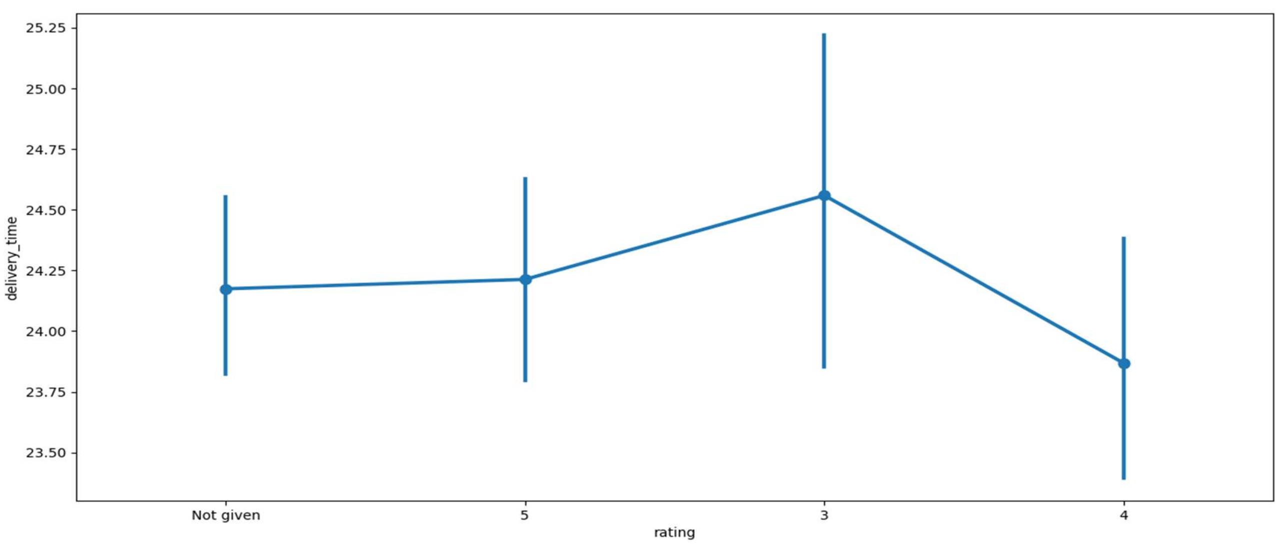
12.3) Day of the week and delivery time relationship



12.4) Run the below code and write your observations on the revenue generated by the restaurants.

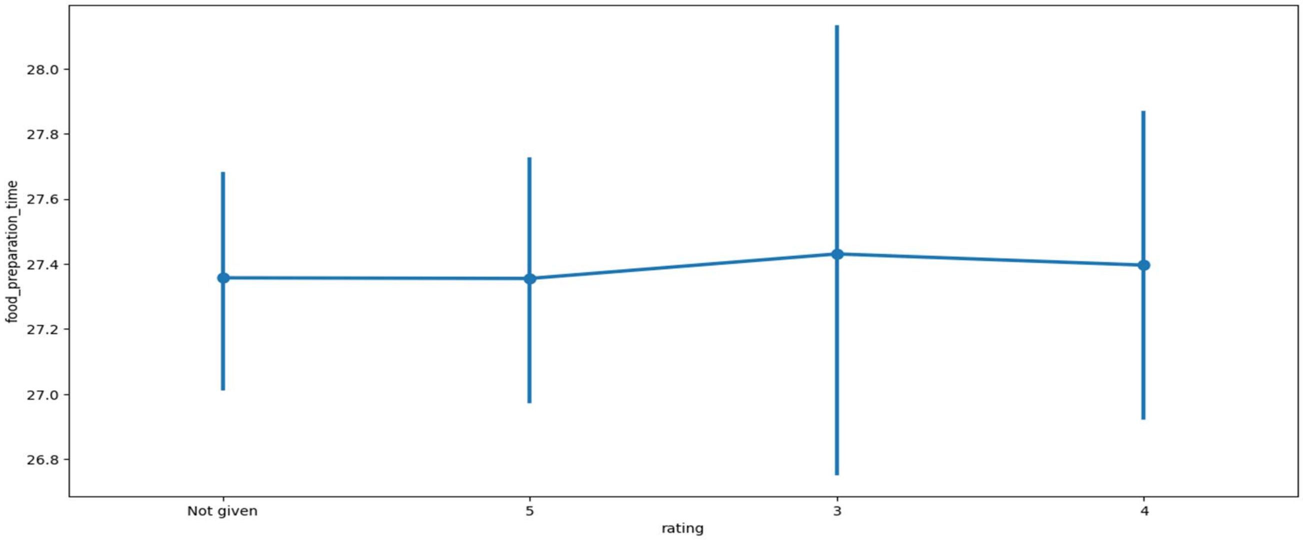
- Shake Shack has the highest order cost at 3,579.53.
- The top three high revenue restaurants are Shake Shack, Meatball Shop, and Blue Ribbon Sushi, with total costs for all orders ranging from about 1,900 to 3,600.

12.5) Rating vs Delivery time relationship

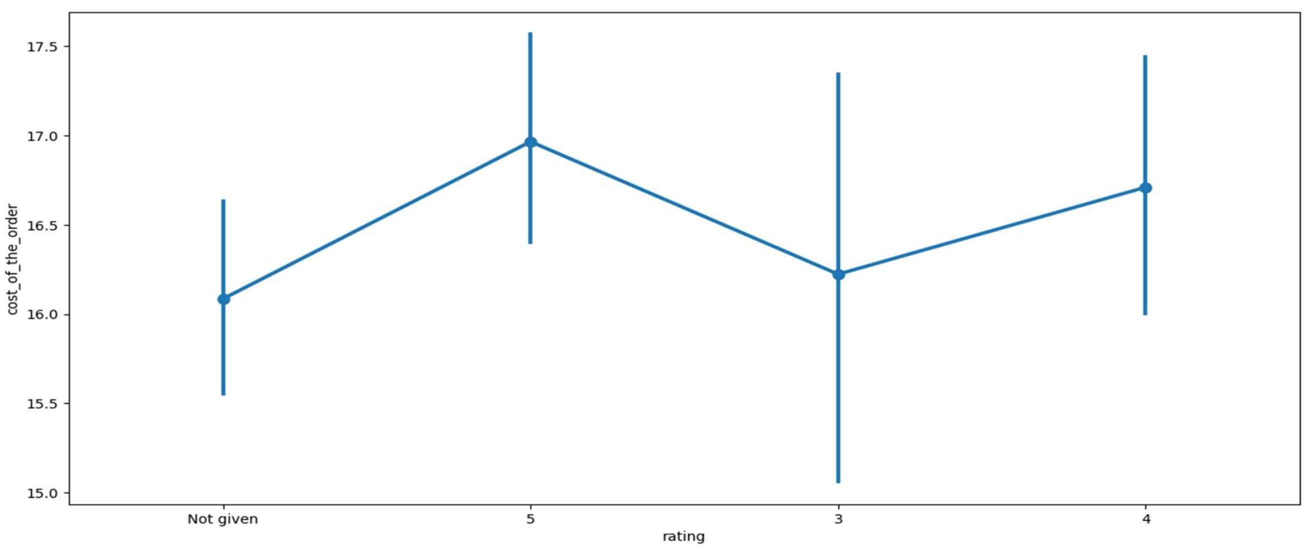


- The data points for ratings 3,4 & 5 are relatively close together.

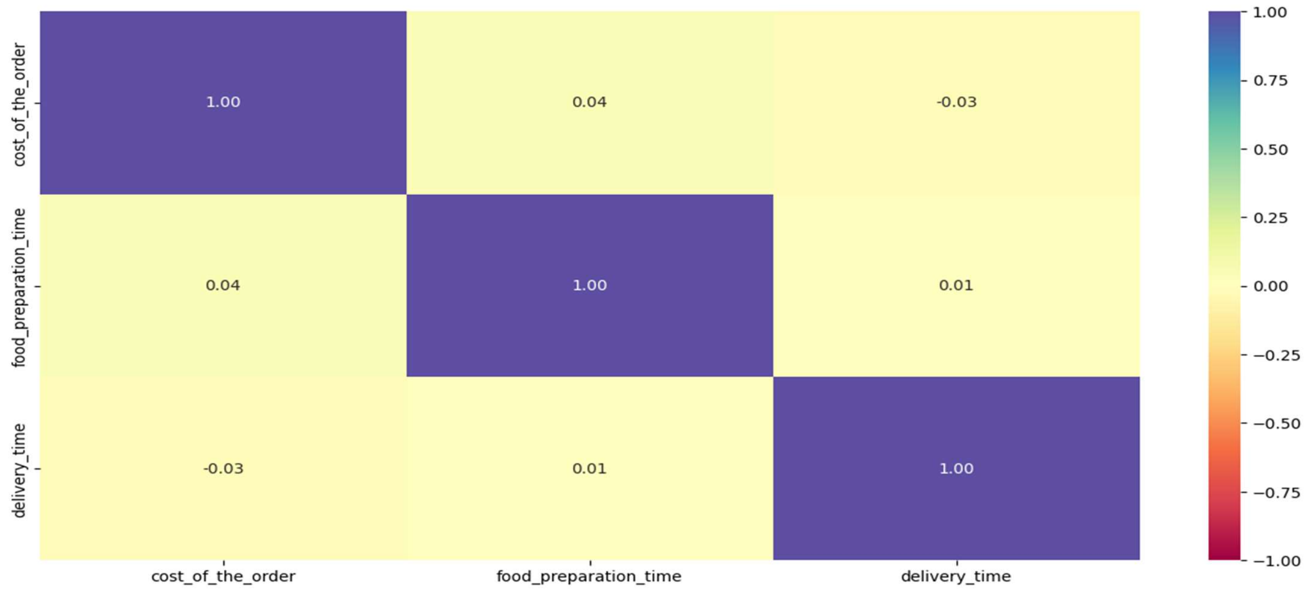
12.6) Rating vs food preparation time



12.7) Rating vs cost of the order relationship



12.8) Correlation among variables



- Cost of the order and food preparation time have a slight corelation. This can be attributed to the quantity of items in each order leading to a higher preparation time.

11) The below restaurants fulfil the criteria-
- Shake shack
- Blue Ribbon Fried Chicken
- The Meatball Shop
- Blue Ribbon Sushi

```
# Filter the rated restaurants
df_rated = df[df['rating'] != 'Not given'].copy()

# Convert rating column from object to integer
df_rated['rating'] = df_rated['rating'].astype('int')

# Create a dataframe that contains the restaurant names with their rating counts
df_rating_count = df_rated.groupby(['restaurant_name'])['rating'].count().sort_values(ascending = False).reset_index()
df_rating_count.head()
```

	restaurant_name	rating
0	Shake Shack	133
1	The Meatball Shop	84
2	Blue Ribbon Sushi	73
3	Blue Ribbon Fried Chicken	64
4	RedFarm Broadway	41

12) The net revenue is around 6166.3 dollars

```
#function to determine the revenue
def compute_rev(x):
    if x > 20:
        return x*0.25
    elif x > 5:
        return x*0.15
    else:
        return x*0

df['Revenue'] = df['cost_of_the_order'].apply(compute_rev) ## Write the apprpriate column name to compute the revenue
df.head()
```

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time	Revenue
0	1477147	337525	Hangawi	Korean	30.75	Weekend	Not given	25	20	7.6875
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	Weekend	Not given	25	23	1.8120
2	1477070	66393	Cafe Habana	Mexican	12.23	Weekday	5	23	28	1.8345
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend	3	25	15	7.3000
4	1478249	76942	Dirty Bird to Go	American	11.59	Weekday	4	25	24	1.7385

```
# get the total revenue and print it
total_rev = df['Revenue'].sum() ## Write the appropriate function to get the total revenue
print('The net revenue is around', round(total_rev, 2), 'dollars')
```

```
The net revenue is around 6166.3 dollars
```

13) Percentage of orders taking more than 60 minutes: 29.24%

```
# Calculate total delivery time and add a new column to the dataframe df to store the total delivery time
df['total_time'] = df['food_preparation_time'] + df['delivery_time']


## Write the code below to find the percentage of orders that have more than 60 minutes of total delivery time (see Question 9 for reference)

print("Percentage of orders above 20 dollars:", round(percentage, 2), '%')
```

```
Percentage of orders above 20 dollars: 29.24 %
```

14) The mean delivery time on weekdays is 28 minutes.

```
# Get the mean delivery time on weekdays and print it
print('The mean delivery time on weekdays is around',
      round(df[df['day_of_the_week'] == 'Weekday']['delivery_time'].mean()),
     'minutes')


## Write the code below to get the mean delivery time on weekends and print it
```

```
The mean delivery time on weekdays is around 28 minutes
```

<u>Actionable Insights:</u>

1) **<u>Revenue</u>**

- The company generated a total revenue of $6,166.30 across all orders.

2) **<u>Delivery</u>**

- The mean delivery time on weekdays is 28 minutes.
- The mean delivery time on weekends is shorter, at 22 minutes.
- 10.54% of orders take more than 60 minutes to be delivered from the time the order is placed.

3) **<u>Customer rating</u>**

- The average customer rating exclude not given is 4.34.

4) **<u>Restaurant Performance</u>**

- Shake Shack is the most popular restaurant with 219 orders and the highest revenue at $703.61.
- Other top-performing restaurants include The Meatball Shop.

Recommendations:

- Focus on reducing delivery times on weekdays to match or exceed weekend performance. Address any systemic issues causing delays.
- Start collecting customer information like customer profiles, loyalty programs or optional surveys during checkout.

- Examine how different demographic groups interact with the platform, including their order preferences, average spend, and ratings.
- Easy way of feedback and do rating process easy
- Focus om reducing then delivery on weekdays.

Thanks & Regards

Hans